Yixiao Wang

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Summary

Incoming PhD student in Computer Science at Duke University with 2+ years of machine learning research experience. Strong background in scalable algorithms, optimization, deep learning, statistical modeling, and discrete optimization. Skilled at turning mathematical insights into efficient, production-ready implementations in C++ and Python. First-author publications at NeurIPS, ICML, and ICLR, with additional conference and workshop papers.

Education

- **Duke University** — *Aug. 2024 – May 2026*
  M.S. in Statistical Science | *Incoming Ph.D. Student in Computer Science* — GPA: 3.95/4.00
- **University of Science and Technology of China, School of the Gifted Young** — *Sep. 2020 – Jul. 2024*
  B.S. in Mathematics — **Top 10%, Outstanding Graduate**

Research Experience

- **Duke University, Interpretable Machine Learning Lab** — *Jun. 2025 – Present*
  *Supervisor: **Prof. Cynthia Rudin*** — Durham, NC, United States
  - **Developed near-optimal algorithms for sparse regression trees in seconds**, unifying constant-leaf and linear-leaf formulations under a dynamic-programming lookahead framework.
  - **Established computational complexity bounds** and **proved approximation guarantees**, showing that the proposed tree-optimization algorithms achieve solutions provably close to the global optimum.
  - **Built** an optimized **C++/Python package** with reproducible benchmarks on 10 UCI datasets, demonstrating consistent order-of-magnitude speedups over optimal baselines (`OSRT`, `STreeD`) while preserving accuracy.
- **Duke University, Department of Computer Science** — *Jun. 2025 – Present*
  *Supervisor: **Prof. Anru Zhang*** — Durham, NC, United States
  - **Co-first-author, NeurIPS 2025 Workshop on What Can('t) Transformers Do? (Oral):** Established the first rigorous in-context learning theory for time series forecasting, showing that Transformers are fundamentally suboptimal: proved a strictly positive excess-risk gap over linear predictors, quantified it with an inverse-linear convergence rate, and demonstrated exponential error compounding under Chain-of-Thought rollout.
  - **Co-first-author, ICLR 2026**: Modeled hidden-state trajectories of Transformers using Menger curvature to analyze logical consistency, uncovering geometric invariants that govern reasoning behavior.
- **Duke Center for Computational Evolutionary Intelligence (CEI)** — *Jan. 2025 – Present*
  *Supervisor: **Prof. Yiran Chen*** — Durham, NC, United States
  - **Developed ZEUS**, a controllable training-free diffusion acceleration framework that achieves up to **3.2× end-to-end speedups** (2.0×–3.2× across SD-2, SDXL, and Flux) while improving perceptual fidelity (**LPIPS $\leq$ 0.08, FID $\leq$ 3.5**) over prior training-free methods.
  - **Co-first-author, ICML 2025**: Developed **SADA**, a stability-guided, training-free adaptive diffusion accelerator that achieves $\geq$1.8× generation speedups across SD-2, SDXL, and Flux, while maintaining perceptual quality (**LPIPS $\leq$ 0.10, FID $\leq$ 4.5**).
  - **Contributing author, AAAI 2026**: Provided theoretical analysis for **FlashSVD**, deriving memory–compute complexity bounds and characterizing rank-dependent speedup and compression thresholds for low-rank Transformer inference.
- **Wake Forest University, Department of Computer Science** — *Jan. 2025 – Aug. 2025*
  *Supervisor: **Prof. Aditya Devarakonda*** — Winston-Salem, NC, United States
  - **Co-first-author NeurIPS 2025**: Proposed and led the development of **Enhanced Cyclic Coordinate Descent (ECCD)**, a Taylor-corrected block coordinate descent method for generalized linear models. The algorithm reduces repeated link-function evaluations and achieves provable improvements in computational efficiency and error bounds.
  - **Developed** a high-performance **C++-based R package** with optimized single-point and pathwise λ solvers, supporting large-scale regularized regression with strong numerical stability.
  - **Demonstrated 2–4× speedups** over `GLMNet` while maintaining convergence and accuracy, validated through extensive benchmarks against state-of-the-art R/Python implementations.

Publications and Preprints

C=Conference, P=Preprint, S=Submitted

[C.1] **Yixiao Wang***, Hayden McTavish*, Varun Babbar*, Margo Seltzer, Cynthia Rudin.
"CLARITree: Cholesky and Lookahead Accelerations for Regression with Interpretable Piecewise Linear Trees," *ICML* 2026. [Paper] [Code]

[C.2] Yufa Zhou*, **Yixiao Wang***, Xunjian Yin*, Shuyan Zhou, Anru R. Zhang.
"The Geometry of Reasoning: Flowing Logics in Representation Space," *ICLR* 2026. [Paper] [Code] [Dataset]

[C.3] **Yixiao Wang***, Zishan Shao*, Ting Jiang, Aditya Devarakonda.
"Enhanced Cyclic Coordinate Descent Methods for Elastic Net Penalized Linear Models," *NeurIPS* 2025. [Paper] [Code]

[C.4] Ting Jiang*, **Yixiao Wang***, Hancheng Ye*, Zishan Shao, Jingwei Sun, Jingyang Zhang, Zekai Chen, Jianyi Zhang, Yiran Chen, Hai Li. **"SADA: Stability-guided Adaptive Diffusion Acceleration,"** *ICML* 2025. [Paper] [Code] [Page] [Slides]

[C.5] Yufa Zhou*, **Yixiao Wang***, Surbhi Goel, Anru R. Zhang. **"Why Do Transformers Fail to Forecast Time Series In-Context?,"** *NeurIPS Workshop on What Can('t) Transformers Do?* 2025 **(Oral Presentation)**. [Paper] [Code]

[C.6] Zishan Shao, Lixun Zhang, Kangning Cui, **Yixiao Wang**, Ting Jiang, Hancheng Ye, Qinsi Wang, Zhixu Du, Yuzhe Fu, Fan Yang, Danyang Zhuo, Yiran Chen, Hai Helen Li, **"DecodeShare: Tracing the Shared Pathways of LLM Decode-Time Decisions,"** *ICML* ***Spotlight*** 2026.

[C.7] Zishan Shao, **Yixiao Wang**, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. **"FlashSVD: Memory-Efficient Inference with Streaming for Low-Rank Models,"** *AAAI* 2026. [Paper] [Code]

[P.1] **Yixiao Wang***, Ting Jiang*, Zishan Shao*, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Jianyi Zhang, Yiran Chen, Hai Li, **"Accelerating Denoising Generative Models is as Easy as Predicting Second-Order Difference,"**. [Paper][Code] [Page]

* denotes co-first authorship; first-/co-first-authored works listed first.

Honors and Awards

- **Centennial & Graduate Professional Fellowship (nominated fellowship, Duke University)** *2026*
- **Outstanding Graduate, University of Science and Technology of China (USTC)** *2024*
- **China Petroleum Scholarship (3 awardees, School of the Gifted Young, USTC)** *2023*
- **Excellent Teaching Assistant (Top 3 in the University, USTC)** *2023*
- **Silver Prize, Outstanding Student Scholarship (Top 15%, USTC)** *2022 & 2021*
- **"Qiang Wei Feng Gong De Yu" (Diligence and Moral Conduct) Scholarship (USTC)** *2022*
- **Excellent President of Student Club (Origami Club, USTC)** *2022*
- **First Prize, Chinese Mathematical Competition for University Students (Top 1%)** *2022 & 2021*
- **Bronze Prize, Outstanding Freshmen Scholarship (Top 35%, USTC)** *2020*

Service

- **Teaching Assistant: Linear Algebra B1 (Spring 2023, Top 3 University-wide)**
- **Reviewer: NeurIPS 2026; AISTATS 2026; ICML 2026; AAAI Workshop 2026**
- **Research Mentorship**
  Ting Jiang (Duke undergraduate, now PhD student at Carnegie Mellon University) Jan 2025 – Sep 2025
- **Leadership: President (Origami Club in USTC)**
- **Volunteer: Hefei Chunyu Parent Support Center, 2021–2022**

Selected Course Projects

- **Airbnb Price Prediction in New York City** (CS 671 Theory & Algorithms of Machine Learning, Duke University) – Achieved top 5 out of 137 participants using advanced stacking strategies, combining XGBoost, LightGBM, and 20+ weak learners for robust generalization. [Report][Code]

Additional Information

Languages: English, Mandarin, Sichuanese dialect.
Programming & Software: Python (primary), C/C++, LaTeX, Linux, HTML/CSS, MATLAB, R.
Interests: Origami and other visual arts (selected works available at portfolio link).